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<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Simple Integral Calculator</title>

  <script src="https://cdnjs.cloudflare.com/ajax/libs/mathjs/12.4.2/math.min.js"></script>

  <style>

    body { font-family: sans-serif; display: flex; justify-content: center; padding: 50px; background:
    #f4f4f9; }

    .card { background: white; padding: 2rem; border-radius: 8px; box-shadow: 0 2px 10px
    rgba(0,0,0,0.1); width: 350px; }

    input { width: 100%; padding: 10px; margin: 10px 0; border: 1px solid #ccc; border-radius: 4px;
    box-sizing: border-box; }

    button { width: 100%; padding: 10px; background: #007bff; color: white; border: none; border-
    radius: 4px; cursor: pointer; }

    button:hover { background: #0056b3; }

    #result { margin-top: 20px; font-weight: bold; color: #333; }

  </style>
</head>

<body>

<div class="card">

  <h3>Definite Integral Calculator</h3>

  <label>Function f(x):</label>

  <input type="text" id="fn" placeholder="e.g., x^2 + sin(x)">

  <label>Lower Limit (a):</label>

  <input type="number" id="lower" value="0">

  <label>Upper Limit (b):</label>

  <input type="number" id="upper" value="1">

  <button onclick="calculate()">Calculate</button>

  <div id="result"></div>

</div>
```

```
<script>

function calculate() {

    const expression = document.getElementById('fn').value;
    const a = parseFloat(document.getElementById('lower').value);
    const b = parseFloat(document.getElementById('upper').value);
    const resultDiv = document.getElementById('result');

    try {

        // Compile the expression

        const code = math.compile(expression);

        // Define the integral using numerical integration (math.integrate is not built-in,
        // so we use a simple trapz rule approximation for this standalone demo)

        const steps = 1000;
        const h = (b - a) / steps;
        let sum = 0;
        for (let i = 0; i < steps; i++) {
            let x = a + i * h;
            sum += code.evaluate({x: x});
        }
        const result = sum * h;

        resultDiv.innerHTML = "Result: " + result.toFixed(6);
        resultDiv.style.color = "#333";
    } catch (err) {

        resultDiv.innerHTML = "Error: Check your syntax.";
        resultDiv.style.color = "red";
    }
}

</script>
```

</body>

</html>